

Conglomerate Mergers: Effects on Growth and Competition

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Abstract

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1 Introduction

[3]

2 Background

Conventional economic models had great success explaining the formation of vertical and horizontal mergers. Despite this, conglomerative mergers have escaped such reasoning so far. Many explanations are based on cost sharing arguments (source), whereby firms split up expenses amongst themselves. One point often made against such an explanation is that this is in effect zero-sum (source), given that one firm's gain is another's loss. However, this critique only holds under the assumption of additive growth. In a setting of multiplicative dynamics, where the next state of some variable (e.g. market share, reputation, size) is dependent on the previous one, the objective is to maximize the growth rate g , the geometric average of periodic increments $\delta_t \in (0, \infty)$, sometimes also referred to as the Compound Annual Growth Rate in investment circles

$$g = \left(\prod_{t=1}^T \delta_t \right)^{\frac{1}{T}} \quad (1)$$

These increments are the changes to the variables in question, such as the change in sales, the effect of marketing campaigns on the way consumers perceive a firm, etc. Crucially, the lower increments are, the greater their impact on the growth rate. While an increment of $\delta_t = 0.2$ already requires the next one to be $\delta_{t+1} = 1.25$ to compensate it, one of $\delta_t = 0.5$ requires δ_{t+1} to be 2. This asymmetry in influence of losses over gains grows exponentially and thus following a logarithmic function of the increments on the growth rate as seen in Figure 1. From the point of view of avoiding the smaller levels of δ_t or increasing them across the board, sharing cost and even revenues in a conglomerate becomes a positive-sum game, as the reduction in the gain of the more successful firms is overcompensated by the increase in gain of the others. This holds true regardless of whether all firms make profits, or only some. Suppose there are two firms in a conglomerate with increments $\delta_{it} > \delta_{jt}$ and sharing of x , given that $\frac{\partial^2 \log(x)}{\partial^2 x} < 0$, it is clear that $\log(\delta_{it}) - \log(\delta_{it} - x) < \log(\delta_{jt} + x) - \log(\delta_{jt})$.

From example, assuming a multiplicative process consisting of normally distributed increments with mean μ and standard deviation σ , the time average growth rate, the average increment, converges to

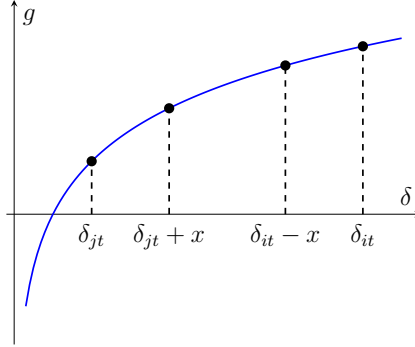


Figure 1: Impact of growth increments on the compound growth rate.

$$g = \mu - \frac{\sigma^2}{2} \quad (2)$$

In addition to the mean μ , the fluctuation of the increments $\frac{\sigma^2}{2}$ has an outsized impact on the growth rate, as the negative fluctuations outweigh the positive ones even though they are symmetric.¹ This dynamic is sometimes referred to as *variance drain* [2] or *volatility drag* [6]. Thus, as outlined, reducing the magnitude of fluctuations is the largest lever to increase the growth rate.

Again, we can link this to sharing, where conglomerate formation increases g for all firms within the entity. Under multiplicative dynamics, sharing can be shown to be growth optimal [4] in the following way. With N firms, whose growth increments δ_i are independently drawn from the same distribution $\mathcal{N} \sim (\mu, \sigma)$, the sharing of their increments increases their individual growth rates such that

$$g = \mu - \frac{\sigma^2}{2N} \quad (3)$$

As the number of firms N in a conglomerate increases, the drag imposed by the fluctuations decreases. In the extreme case of $N \rightarrow \infty$ each firm is able to realize the ensemble average μ with every increment. Of course there are limits to how big a firm and thus a conglomerate can grow, given that transaction cost, especially for coordination increase [1]. Any increase in conglomerate size will increase transaction cost, while at the same time increasing the growth rate for all firms included. Thus, one can see this

¹ $(1 - \delta_t)(1 + \delta_t) = 1 - \delta_t^2 < 1$.

kind of cooperation as a hedge, or an insurance against unfavorable growth increments, the premium for which is the cost of coordination. Clearly, there is a trade-off involved, and a conglomerate size, though certainly varying, which optimizes it.

3 Model

3.1 Random Multiplicative Growth

3.1.1 Growth Process

To investigate how this influences the incentives in conglomerate formation, we set up an Agent Based Model with M markets that are uncorrelated between each other, each containing an amount of N firms. Firms face random multiplicative growth that is unique for their market. Each increment δ_{it} is drawn from a normal distribution $\mathcal{N}(\mu_m, \sigma_m)$ with positive ensemble average μ_m . In this framework each firm's periodic state, which can be seen as its size, is multiplicatively linked to its previous one.

$$s_{it+1} = s_{it}\delta_{it} \quad \text{with} \quad \delta_t \sim \mathcal{N}(\mu_m, \sigma_m) \quad (4)$$

We model markets such that mean and the standard deviation of the distribution over their growth increments are positively correlated, meaning that a higher mean μ_m comes with a higher standard deviation σ_m . As explained in section 2, in multiplicative processes one is not just maximizing for the ensemble average μ_m , but for the growth rate g . Hence, a higher scale σ_m decreases the growth rate, through the volatility tax.

Each firm thus realizes a gain of $\Delta_{it} = s_{it-1}\delta_{it} - s_{it-1}$. For any firm not part of a conglomerate this will also be their period profit $\Pi_{it} = \Delta_{it}$. Firms in conglomerates C pool an amount $\alpha\Delta_{it}$, which flows into the conglomerate's period pool $\Omega_{ct} = \sum_{i \in c} \alpha\Delta_{it}$. However, as conglomerates grow in size, maintaining them becomes ever more difficult as transaction costs increase. We model these as a logistic function of the form

$$\Phi(O_c, M) = \frac{1}{1 + \exp^{-|O_c| + 0.4M}} \quad (5)$$

This factor scales the cost according to conglomerate size $\sum_{i \in c} s_{it}$ such that the pool size is given by

$$\Omega_{ct} = \sum_{i \in c} \alpha \Delta_{it} - \Phi(O_c, M) \sum_{i \in c} s_{it} \quad (6)$$

where the set of markets a conglomerate operates in is defined as $O_c \subseteq M$. Notice that the factor 0.4 in the logistic function scales the costs such that a conglomerate operating in 40 % of all markets would lose 50 % of their market value in transaction cost every round. Lastly, this pool will be shared equally, determining a members profit by $\Pi_{it} = (1 - \alpha)\Delta_{it} + \frac{\Omega_{ct}}{|O_{c_i}|}$. Consequently, any firms next state is given by $s_{it+1} = s_{it} + \Pi_{it}$.

3.1.2 Merging Process

At the beginning of every time step t each firm $i \in M \times N$ gets a chance to choose another with probability p . If successful, it will select a random other firm j , among the remaining $(M \times N) - 1$ firms. We allow conglomerates to only operate one firm per market, meaning it can have a maximum of M members. Hence, conglomeration is only *feasible* if firm j and i :

1. don't operate in the same market ($m_i \neq m_j$)
2. are not part of a conglomerate that operates in a market in which the other firm, or the conglomerate which it is part of operates in ($O_{c_i} \cap O_{c_j} = \emptyset$)

We assume that if both firms are in conglomerates that these will merge as well. After confirming *feasibility*, merging firms determine *desirability*, meaning that they would both benefit from joining forces. They do so by comparing the realized pool Ω size over the last l periods, against a "synthetic" pool $\hat{\Omega}$ size if they had merged. If $\hat{\Omega} > \Omega$ the merger will be implemented.

3.1.3 Splitting Process

After an initial l periods each firm that is part of a conglomerate evaluates its performance within it by comparing its growth rate $g_{i(t-l):t}$ being part of the conglomerate against the potential rate $\hat{g}_{i(t-l):t}$ it had realized, had it remained outside. Upon being worse off, a firm exits the conglomerate and continues to operate alone. The full model can be seen in algorithm 1.

Algorithm 1: Model flow with random growth.

```

for  $t$  in  $T$  do
  foreach  $i \in N \times M$  do
     $\lfloor$  Merge with  $j$  if  $(m_i \neq m_j)$  and  $(O_{c_i} \cap O_{c_j} = \emptyset)$  and  $\hat{\Omega} > \Omega$ 
  foreach  $c \in C$  do
     $\Omega_{ct} = \sum_{i \in c} \alpha \Delta_{it} - \Phi(O_c, M) \sum_{i \in c} s_{it}$ 
    foreach  $i \in c$  do
       $\lfloor \Pi_{it} = (1 - \alpha) \Delta_{it} + \frac{\Omega_{ct}}{|O_c|}$ 
    foreach  $i \notin c$  for all  $c \in C$  do
       $\lfloor \Pi_{it} = \Delta_{it}$ 
    foreach  $i \in c$  for all  $c \in C$  do
       $\lfloor$  Exit  $c$  if  $\hat{g}_{i(t-l):t} > g_{i(t-l):t}$ 

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3.2 Endogenous Pricing

Going beyond random growth, we enable our firms to optimize revenues by setting their period prices p_{it} for which they receive the following demand

$$D_{it}(\mathbf{P}_t, \theta_{it}) = 1 - (1 - \theta_t)p_{it} + 0.5 \max_{j:j \neq i} \mathbf{P}_t \quad (7)$$

where θ_{it} is the firm's current market share, which is the relative cumulative profits within a market

$$\theta_{it} = \frac{\sum_{\tau=1}^t \Pi_{i\tau}}{\sum_{\tau=1}^t \sum_{j:j \neq i} \Pi_{j\tau}} \quad (8)$$

Consequently, period revenues take the form of $D_{it}(\mathbf{P}_t, \theta_{it})p_{it}$. We further assume stochastic costs γ that are drawn from a log-normal distribution $\mathcal{L}(0, 1)$ that is scaled to match the mean growth rate of revenues per period. Hence, a firm's pre-pooling profit is given by $\Delta_{it} = D_{it}(\mathbf{P}_t, \theta_{it})p_{it} - \gamma$. If the firm is not in a conglomerate, there is no pooling, meaning $\Pi_{it} = \Delta_{it}$. In case it is part of a conglomerate, it will continue as outlined in section 3.1.

To learn the optimal pricing policy π , the optimal response given a set of information, we trained a reinforcement learning agent using the Proximal Policy Optimization algorithm [5]. We trained the agents in a separate environment being tasked to only optimize revenues $D_{it}(\mathbf{P}_t, \theta_{it})p_{it}$. Its information set contains its price and market share in the previous round, as well as the maximum price of its competitors in the previous two rounds,

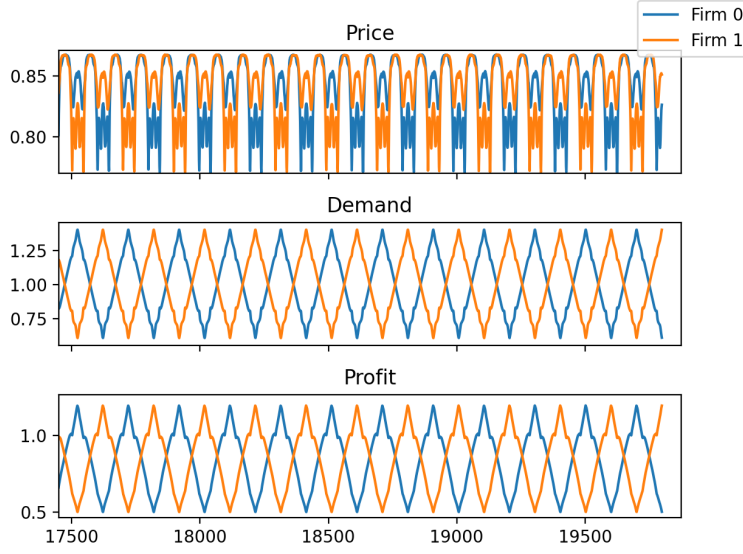


Figure 2: Testing a trained policy with countervailing market shares.

$\{p_{it-1}, \theta_{it-1}, \max_{j:j \neq i} \mathbf{P}_{jt-1}, \max_{j:j \neq i} \mathbf{P}_{jt-2}\}$. To expose the agents to possible market share constellations, we exogenously set the θ_{it} . The resulting policy can be seen in figure 2. By combining all these elements, the resulting model flow can be seen in algorithm 2.

Algorithm 2: Model flow with endogenous pricing.

```

for  $t$  in  $T$  do
  foreach  $i \in N \times M$  do
     $\lfloor$  Merge with  $j$  if  $(m_i \neq m_j)$  and  $(O_{c_i} \cap O_{c_j} = \emptyset)$  and  $\hat{\Omega} > \Omega$ 
  foreach  $i \in N \times M$  do
     $\lfloor$   $p_{it} = \pi(p_{it-1}, \theta_{it-1}, \max_{j:j \neq i} \mathbf{P}_{jt-1}, \max_{j:j \neq i} \mathbf{P}_{jt-2})$ 
   $\Delta_t = D(\mathbf{P}_t, \theta_t) \mathbf{P}_t - \gamma$ 
  foreach  $c \in C$  do
     $\lfloor$   $\Omega_{ct} = \sum_{i \in c} \alpha \Delta_{it} - \Phi(O_c, M) \sum_{i \in c} s_{it}$ 
    foreach  $i \in c$  do
       $\lfloor$   $\Pi_{it} = (1 - \alpha) \Delta_{it} + \frac{\Omega_{ct}}{|O_c|}$ 
  foreach  $i \notin c$  for all  $c \in C$  do
     $\lfloor$   $\Pi_{it} = \Delta_{it}$ 
  foreach  $i \in M \times N$  do
     $\lfloor$   $\theta_{it} = \frac{\sum_{\tau=1}^t \Pi_{i\tau}}{\sum_{\tau=1}^t \sum_{j:j \neq i} \Pi_{j\tau}}$ 
  foreach  $i \in c$  for all  $c \in C$  do
     $\lfloor$  Exit  $c$  if  $\hat{g}_{i(t-l):t} > g_{i(t-l):t}$ 

```

4 Results

4.1 Random Multiplicative Growth

We implemented the models outlined in the previous section in python with a parameter specification that can be found in Table 1. Note that we conducted extensive robustness testing, none of which altered the central findings of this section.

Parameter	Value
M	100
N	100
T	1000
l	50
p	0.05
α	$[0, 0.5]$

Table 1: Model Parameters and their values

To answer the questions outlined in section 2, we analyze certain outputs

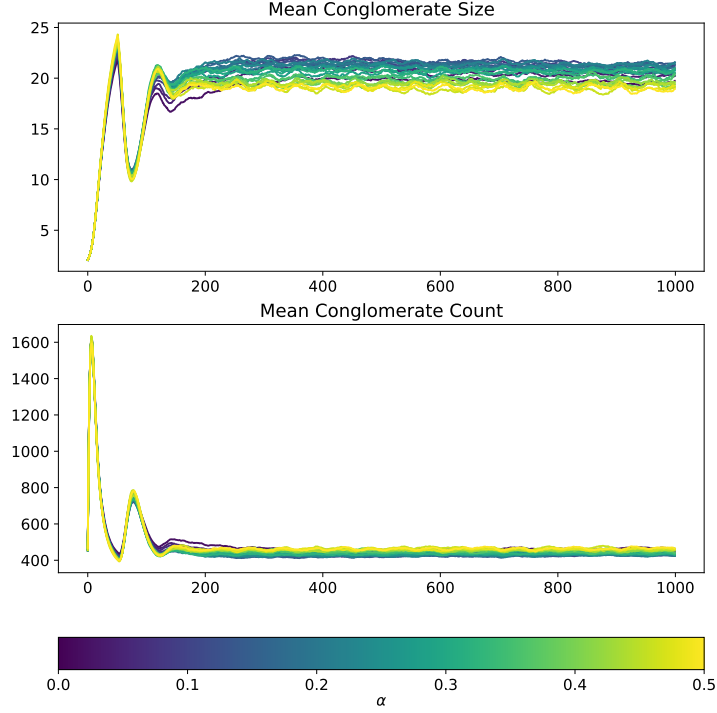


Figure 3: Conglomerate sizes and number of conglomerates.

of our model. We gradually increase the parameter of interest, the sharing rate α , by 0.02 to investigate the behavior of our model. To gain further confidence in our result, we ran each experimental setup (each level of α) 10 times in a Monte Carlo type setting and averaged them. Firstly, Figure 3 indicates a stable tendency to form conglomerates, which after initial fluctuations, partially due the review period l , settles to a stable level up until the final number of iterations T is reached. Both the number of conglomerates, and their size converge to a regular pattern. Interestingly, while initially increasing the sharing rate leads to larger conglomerates, there seems to be an optimum in regard to conglomerate size somewhere between 0.2 and 0.3. This is also reflected, though not as strongly, in the number of conglomerates, which bottoms out at the same level, as expected. Both of these findings, lend support to the idea of collaborative growth being able to explain incentives to merge in a conglomerative manner.

To examine the impact of such mergers on the competitive landscape as well as that transaction cost, we fit a polynomial of degree two to the relation between conglomerate size and average market share of firms within a given conglomerate $\bar{\Omega}_{ct} = \beta_{0t} + \beta_{1t}|O_{ct}| + \beta_{2t}|O_{ct}|^2$ for each step in the

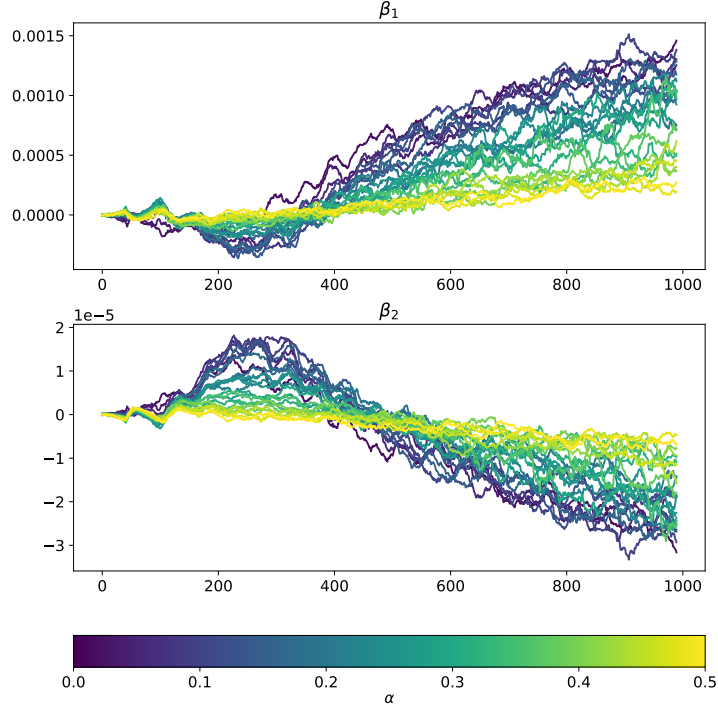


Figure 4: Average estimates for the impact of conglomerate size on market share.

model and display the result in Figure 4. For all levels of α we see that an increase in conglomerate size increases market share ($\beta_{1t} > 0$), once the initial fluctuations pass. However, we also note that β_{2t} is negative in the same region, meaning that growing too large will decrease competitiveness. This sheds light on our question 3, by indicating the existence an upper bound on conglomerate size. Note that, as we increase the sharing rate, these effects become ever more muted, pushing the estimates for both β_{1t} and β_{2t} towards 0. This is a direct consequence of a decrease in the variability and scale of market share inequality.

Figure 5 provides an initial insight into this dynamic by illustrating the market share distribution across different quantiles. The graph displays the cumulative market share attained by each quantile of firms, ordered from largest to smallest. These are taken across every market and averaged across all quantiles for every point in time. In our setup with 100 firms per market a perfectly equal result would assign every firm a market share of 0.01. As expected, the case without pooling (the dashed line) returns the standard result of random multiplicative growth, exhibiting extreme inequality with the largest firm making up almost the entirety of the market and even the 90th

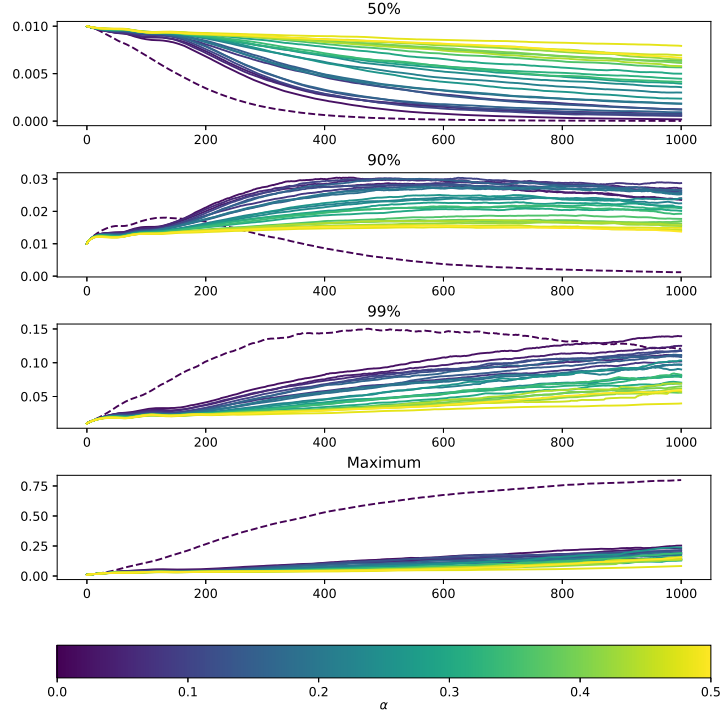


Figure 5: Market share of a given quantile.

percentile quickly dropping to 0% share. Increasing α just ever so slightly already changes this dynamic drastically for the largest firm. Furthermore, all quantiles become more equal in market share until they eventually remain close to equality for the entirety of the simulation. As already alluded to, this also explains why conglomerate size becomes ever less predictive with an increase in sharing as the variation within the market shares becomes ever smaller.

This finding is being confounded by Figure 6, for which we calculate the Gini coefficient for every market, divide these into quantiles over which we can then average across simulations. Firstly, we can see that higher quantiles reach their maximum inequality earlier than lower ones. More importantly though, larger values of α slow the increase and cap the maximum level of inequality for all quantiles. Again, as the sharing rate reaches its maximum inequality as measured by the Gini coefficient is minimized, mirroring finding of the previous analysis.

Yet, these two analyses are both static measures for a distribution, meaning that they can only provide information on the aggregate. To address this, we follow each firm throughout each simulation and tally both the difference

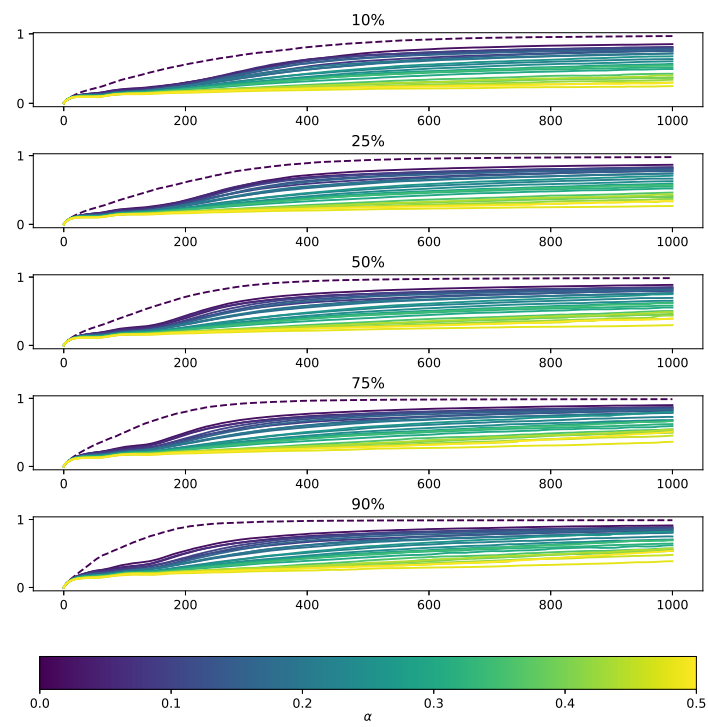


Figure 6: Quantiles of Gini coefficients.

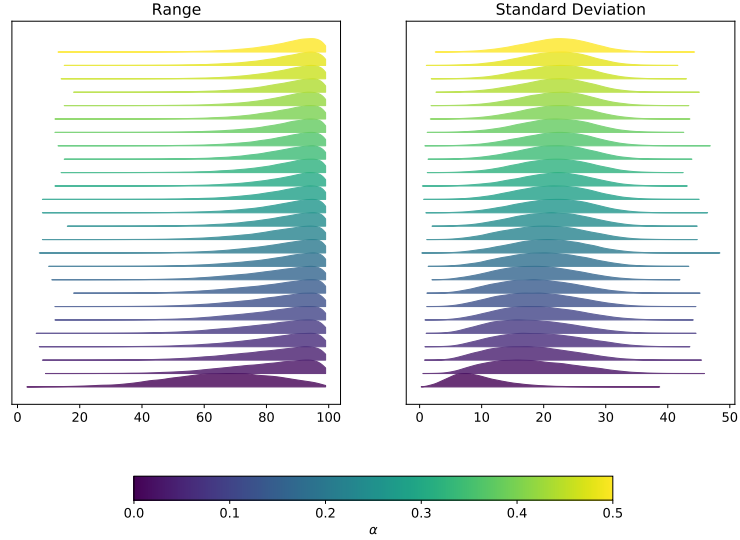


Figure 7: Distribution over ranks.

between its maximum and minimum rank and the standard deviation of its ranks, and plot the resulting distributions over all firms and counterfactuals in Figure 7. Given that there are 100 firms per market, this sets an upper limit on mobility as measured by both metrics. Again we observe that introducing just a small amount of sharing changes the dynamic drastically, as both measures shift completely in their pattern, raising mobility drastically. In addition, increasing the sharing rate increases the variability of ranks, shifting the distribution to the right. Interestingly, as the maximum range is already reached for any sharing level, we only see more mass of the distribution moving to the right.

5 Discussion

6 Conclusion

References

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